

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{r} (f(x_1) + \cdots + f(x_r)).$$

(2) More generally, prove that for any permutation σ and function $f : X \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{p_x} (f(x) + f(\sigma(x)) + \cdots + f(\sigma^{p_x-1}(x)))$$

where the orbit of x has cardinality p_x under σ .

Exercise 2.6.2. Mimic the proof of Lemma 2.27 (or give the details of a deduction) to prove Corollary 2.28.

Exercise 2.6.3. Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system. Prove that, for any $f \in L^1_\mu$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^{-n} x)$$

almost everywhere.

Exercise 2.6.4. Fill in the details to prove the estimate in (2.28).

Exercise 2.6.5. Formulate and prove a pointwise ergodic theorem for a measurable function $f \geq 0$ with $\int f \, d\mu = \infty$, under the assumption of ergodicity.

2.7 Strong-Mixing and Weak-Mixing

In this section we step back from thinking of measure-preserving transformations through the functional-analytic prism of their action on L^p spaces to the more fundamental questions discussed in Sects. 2.2 and 2.3. Namely, if A is a measurable set, what can be said about how the set $T^{-n}A$ is spread around the whole measure space for large n ?

An easy consequence of the mean ergodic theorem is that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if

$$\frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n \xrightarrow{L^2_\mu} \int f \, d\mu$$

as $N \rightarrow \infty$ for every $f \in L^2_\mu$. It follows that $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if

$$\frac{1}{N} \sum_{n=0}^{N-1} \langle f \circ T^n, g \rangle \longrightarrow \int f \, d\mu \int g \, d\mu \quad (2.30)$$

as $N \rightarrow \infty$ for any $f, g \in L^2_\mu$. The characterization in (2.30) can be cast in terms of the behavior of sets to show that $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if

$$\frac{1}{N} \sum_{n=0}^{N-1} \mu(A \cap T^{-n}B) \longrightarrow \mu(A)\mu(B) \quad (2.31)$$

as $N \rightarrow \infty$ for all $A, B \in \mathcal{B}$. One direction is clear: if T is ergodic, then the convergence (2.30) may be applied with $g = \chi_A$ and $f = \chi_B$.

Conversely, if $T^{-1}B = B$ then the convergence (2.31) with $A = X \setminus B$ implies that $\mu(X \setminus B)\mu(B) = 0$, so T is ergodic.

There are several ways in which the convergence (2.31) might take place. Recall that measurable sets in $(X, \mathcal{B}, \mu)$ may be thought of as events in the sense of probability, and events $A, B \in \mathcal{B}$ are called *independent* if

$$\mu(A \cap B) = \mu(A)\mu(B).$$

Clearly if the action of T contrives to make $T^{-n}B$ and A become *independent* in the sense of probability for all large n , then the convergence (2.31) is assured. It turns out that this is too much to ask (see Exercise 2.7.1), but asking for $T^{-n}B$ and A to become *asymptotically independent* leads to the following non-trivial definition.

Definition 2.32. A measure-preserving system $(X, \mathcal{B}, \mu, T)$ is *mixing* if

$$\mu(A \cap T^{-n}B) \longrightarrow \mu(A)\mu(B)$$

as $n \rightarrow \infty$, for all $A, B \in \mathcal{B}$.

Mixing is also sometimes called *strong-mixing*, in contrast to weak-mixing and mild-mixing.

Example 2.33. A circle rotation $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$ is not mixing. There is a sequence $n_j \rightarrow \infty$ for which $n_j\alpha \pmod{1} \rightarrow 0$ (if α is rational we may choose to have $n_j\alpha \pmod{1} = 0$). If $A = B = [0, \frac{1}{2}]$ then $m_{\mathbb{T}}(A \cap R_\alpha^{n_j}A) \rightarrow \frac{1}{2}$, so R_α is not mixing.

It is clear that some measure preserving systems make many sets become asymptotically independent as they move apart in time (that is, under iteration), leading to the following natural definition due to Rokhlin [316].

Definition 2.34. A measure-preserving system $(X, \mathcal{B}, \mu, T)$ is *k-fold mixing*, *mixing of order k* or *mixing on k + 1 sets* if

$$\mu(A_0 \cap T^{-n_1}A_1 \cap \cdots \cap T^{-n_k}A_k) \longrightarrow \mu(A_0) \cdots \mu(A_k)$$

as

$$n_1, n_2 - n_1, n_3 - n_2, \dots, n_k - n_{k-1} \longrightarrow \infty$$

for any sets $A_0, \dots, A_k \in \mathcal{B}$.

Thus mixing coincides with mixing of order 1. One of the outstanding open problems in classical ergodic theory is that it is not known⁽²⁵⁾ if mixing implies mixing of order k for every $k \geq 1$.

Despite the natural definition, mixing turns out to be a rather special property, less useful and less prevalent than a slightly weaker property called weak-mixing introduced by Koopman and von Neumann [209]⁽²⁶⁾. Nonetheless, many natural examples are mixing of all orders (see the argument in Proposition 2.15 and Exercise 2.7.9 for example).

Definition 2.35. A measure-preserving system $(X, \mathcal{B}, \mu, T)$ is *weak-mixing* if

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)| \longrightarrow 0$$

as $N \rightarrow \infty$, for all $A, B \in \mathcal{B}$.

Notice that for any sequence (a_n) ,

$$\lim_{n \rightarrow \infty} a_n = 0 \implies \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n |a_i| = 0,$$

but the converse does not hold because the second property permits $|a_n|$ to be large along an infinite but thin set of values of n . Thus at the level simply of sequences, weak-mixing seems to be strictly weaker than strong-mixing. It turns out that this is also true for measure-preserving transformations—there are weak-mixing transformations that are not mixing⁽²⁷⁾.

Weak-mixing and its generalizations will turn out to be central to Furstenberg's proof of Szemerédi's theorem presented in Chap. 7. The first intimation that weak-mixing is a natural property comes from the fact that it has many equivalent formulations, and we will start to define and explore some of these in Theorem 2.36 below.

For one of these equivalent properties, it will be useful to recall some terminology concerning the operator U_T on the Hilbert space L^2_μ associated to a measure-preserving transformation T of $(X, \mathcal{B}, \mu)$. An *eigenvalue* is a number $\lambda \in \mathbb{C}$ for which there is an *eigenfunction* $f \in L^2_\mu$ with $U_T f = \lambda f$ almost everywhere. Notice that 1 is always an eigenvalue, since a constant function f will satisfy $U_T f = f$. Any eigenvalue λ lies on $\mathbb{S}^1$, since U_T is an isometry of L^2_μ . A measure-preserving transformation T is said to have *continuous spectrum* if the only eigenvalue of T is 1 and the only eigenfunctions are the constant functions.

Recall that a set $J \subseteq \mathbb{N}$ is said to have *density*

$$\mathbf{d}(J) = \lim_{n \rightarrow \infty} \frac{1}{n} |\{j \in J \mid 1 \leq j \leq n\}|$$

if the limit exists.

Theorem 2.36. *The following properties of a system $(X, \mathcal{B}, \mu, T)$ are equivalent.*

- (1) T is weakly mixing.
- (2) $T \times T$ is ergodic with respect to $\mu \times \mu$.
- (3) $T \times T$ is weakly mixing with respect to $\mu \times \mu$.
- (4) For any ergodic measure-preserving system $(Y, \mathcal{B}_Y, \nu, S)$, the system

$$(X \times Y, \mathcal{B} \otimes \mathcal{B}_Y, \mu \times \nu, T \times S)$$

is ergodic.

- (5) The associated operator U_T has no non-constant measurable eigenfunctions (that is, T has continuous spectrum).
- (6) For every $A, B \in \mathcal{B}$, there is a set $J_{A,B} \subseteq \mathbb{N}$ with density zero for which

$$\mu(A \cap T^{-n}B) \rightarrow \mu(A)\mu(B)$$

as $n \rightarrow \infty$ with $n \notin J_{A,B}$.

- (7) For every $A, B \in \mathcal{B}$,

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)|^2 \rightarrow 0$$

as $N \rightarrow \infty$.

The proof of Theorem 2.36 will be given in Sect. 2.8.

Corollary 2.37. *If $(X, \mathcal{B}_X, \mu, T)$ and $(Y, \mathcal{B}_Y, \nu, S)$ are both weak-mixing, then the product system $(X \times Y, \mathcal{B} \otimes \mathcal{C}, \mu \times \nu, T \times S)$ is weak-mixing.*

Corollary 2.38. *If T is weak-mixing, then for any k the k -fold Cartesian product $T \times \cdots \times T$ is weak-mixing with respect to $\mu \times \cdots \times \mu$.*

Corollary 2.39. *If T is weak-mixing, then for any $n \geq 1$, the n th iterate T^n is weak-mixing.*

Example 2.40. We know that the circle rotation $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$ defined by

$$R_\alpha(t) = t + \alpha \pmod{1}$$

is not mixing, but is ergodic if $\alpha \notin \mathbb{Q}$ (cf. Proposition 2.16 and Example 2.33). It is also not weak-mixing; this may be seen using Theorem 2.36(2) since the function $(x, y) \mapsto e^{2\pi i(x-y)}$ from $\mathbb{T} \times \mathbb{T} \rightarrow \mathbb{S}^1$ is a non-constant function preserved by $R_\alpha \times R_\alpha$.

Exercises for Sect. 2.7

Exercise 2.7.1. Show that if a measure-preserving system $(X, \mathcal{B}, \mu, T)$ has the property that for any $A, B \in \mathcal{B}$ there exists N such that

$$\mu(A \cap T^{-n}B) = \mu(A)\mu(B)$$

for all $n \geq N$, then it is trivial in the sense that $\mu(A) = 0$ or 1 for every $A \in \mathcal{B}$.

Exercise 2.7.2. ⁽²⁸⁾ Show that if a measure-preserving system $(X, \mathcal{B}, \mu, T)$ has the property that

$$\mu(A \cap T^{-n}B) \rightarrow \mu(A)\mu(B)$$

uniformly as $n \rightarrow \infty$ for every measurable $A \subseteq B \in \mathcal{B}$, then it is trivial in the sense that $\mu(A) = 0$ or 1 for every $A \in \mathcal{B}$.

Exercise 2.7.3. This exercise generalizes the argument used in the proof of Proposition 2.15 and relates to the material in Appendix A. A collection $\mathcal{A}$ of measurable sets in $(X, \mathcal{B}, \mu)$ is called a *semi-algebra* (cf. Appendix A) if

- $\mathcal{A}$ contains the empty set;
- for any $A \in \mathcal{A}$, $X \setminus A$ is a finite union of pairwise disjoint members of $\mathcal{A}$;
- for any $A_1, \dots, A_r \in \mathcal{A}$, $A_1 \cap \dots \cap A_r \in \mathcal{A}$.

The smallest σ -algebra containing $\mathcal{A}$ is called the σ -algebra generated by $\mathcal{A}$. Assume that $\mathcal{A}$ is a semi-algebra that generates $\mathcal{B}$, and prove the following characterizations of the basic mixing properties for a measure-preserving system $(X, \mathcal{B}, \mu, T)$:

(1) T is mixing if and only if

$$\mu(A \cap T^{-n}B) \longrightarrow \mu(A)\mu(B)$$

as $n \rightarrow \infty$ for all $A, B \in \mathcal{A}$.

(2) T is weak-mixing if and only if

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)| \longrightarrow 0$$

as $N \rightarrow \infty$ for all $A, B \in \mathcal{A}$.

(3) T is ergodic if and only if

$$\frac{1}{N} \sum_{n=0}^{N-1} \mu(A \cap T^{-n}B) \longrightarrow \mu(A)\mu(B)$$

as $N \rightarrow \infty$ for all $A, B \in \mathcal{A}$.

Exercise 2.7.4. Let $\mathcal{A}$ be a generating semi-algebra in $\mathcal{B}$ (cf. Exercise 2.7.3), and assume that for $A \in \mathcal{A}$, $\mu(A \Delta T^{-1}A) = 0$ implies $\mu(A) = 0$ or 1. Does it follow that T is ergodic?

Exercise 2.7.5. Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is mixing if and only if

$$\lim_{n \rightarrow \infty} \langle U_T^n f, g \rangle = \langle f, 1 \rangle \cdot \langle 1, g \rangle$$

for all f and g lying in a dense subset of L_μ^2 .

Exercise 2.7.6. Use Exercise 2.7.5 and the technique from Theorem 2.19 to prove the following.

- (1) An ergodic automorphism of a compact abelian group is mixing with respect to Haar measure.
- (2) An ergodic automorphism of a compact abelian group is mixing of all orders with respect to Haar measure.

Exercise 2.7.7. Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is weak-mixing if and only if

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |\langle U_T^n f, g \rangle - \langle f, 1 \rangle \cdot \langle 1, g \rangle| = 0$$

for any $f, g \in L_\mu^2$

Exercise 2.7.8. Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is weak-mixing if and only if

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |\langle U_T^n f, f \rangle - \langle f, 1 \rangle \cdot \langle 1, f \rangle| = 0$$

for any $f \in L_\mu^2$.

Exercise 2.7.9. Show that a Bernoulli shift (cf. Example 2.9) is mixing of order k for every $k \geq 1$.

Exercise 2.7.10. Prove the following result due to Rényi [308]: a measure-preserving transformation T is mixing if and only if

$$\mu(A \cap T^{-n}A) \rightarrow \mu(A)^2$$

for all $A \in \mathcal{B}$. Deduce that T is mixing if and only if $\langle U_T^n f, f \rangle \rightarrow 0$ as $n \rightarrow \infty$ for all f in a set of functions dense in the set of all L^2 functions of zero integral.

Exercise 2.7.11. Prove that a measure-preserving transformation T is weak-mixing if and only if for any measurable sets A, B, C with positive measure, there exists some $n \geq 1$ such that $T^{-n}A \cap B \neq \emptyset$ and $T^{-n}A \cap C \neq \emptyset$. (This is a result due to Furstenberg.)